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Enthalpies of Solution, Partial Molal Heat Capacities and Apparent Molal Volumes of Sugars and Polyols in Water

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Integral enthalpies of solution of some sugars and polyols in water at low concentrations have been determined calorimetrically at 25 and 35°C. These data have been used to derive heat capacities of solution ΔC_p^o at 30°C. Partial molal heat capacities $\overline{C}_{p,2}^o$ have been obtained by combining ΔC_p^o with $C_{p,2}^*$, the heat capacity of pure solid compounds. Apparent molal volumes have been obtained from density data. The sugars as well as polyols show significantly high positive ΔC_p^o and $\overline{C}_{p,2}^o$ values. The results have been explained in terms of a specific hydration model. The effect of substitution of -OH by glycosidic -OCH_3 and of -CHOH by deoxy-CH_2 are also discussed.

KEY WORDS: Enthalpy of solution; partial molal heat capacity; density; apparent molal volume; sugars; polyols; specific hydration; hydrophobic hydration.

1. INTRODUCTION

In view of increasing biological and technological applications,[1,2] the hydration of sugars is a topic of considerable importance. Spectroscopic[3] and volumetric[4] studies by Franks, Suggett and their collaborators on the hydration of monosaccharides have invoked the concept of solvent structure compatibility to explain the observed differences in solution behavior of polyhydroxy molecules which differ only in the positions and orientations of potential hydration sites. However, a comprehensive understanding of the mode of hydration is yet to emerge. We felt that a systematic and extensive study of heat capacities and partial molal volumes could contribute to a better under-

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standing of the mode of hydration of sugars and, in particular, could establish a possible correlation between the stereochemistry and hydration properties of the solute.

We report calorimetrically determined integral enthalpies of solution (at low concentrations) at 25 and 35°C and partial molal volumes at 25°C for a series of monosaccharides, deoxymonosaccharides, methylglycosides, and polyols.

2. EXPERIMENTAL

2.1. MATERIALS

Sugars and polyols (Sigma and P. L. Biochemicals) of highest purity grade were dried under vacuum at 60°C to constant weight and used without further purification. Water was purified by passing distilled water through Barnstead mixed-bed ion-exchange resin. Conductivity water, prepared by distilling the deionized water from alkaline permanganate, was used for density measurements.

2.2. CALORIMETER

The isoperibol submarine calorimeter employed for measuring the enthalpies of solution is similar to that described earlier.[5] It consists of a glass dewar (capacity 550 ml), immersed in a water bath regulated to ± 0.0005°C by a proportional temperature controller (Tronac Model PTC 40). A 3000 ohm thermistor (Y.S.I. Ohio) was the temperature sensing probe. Output voltage from the bridge was amplified by a nanovolt DC amplifier (Keithley Model 140) and recorded. The calorimeter was calibrated electrically immediately before and after each experiment. A constant current source (Keithley Model 227) coupled with a digital timer (Thermadyne, New Delhi, Model DE 203) was used as a power source for the calibration heater.

Resolution of the temperature measurement system over the whole range was 1×10^{-5}°C . The operation of the calorimeter was checked by measuring the enthalpies of solution of KCl in water and of THAM (tris-hydroxymethylaminomethane) in 0.1*N* HCl. ΔH_s° for KCl at 25°C agree within 1% with reported values.[6,7] The ΔH_s° of THAM in 0.1*N* HCl agreed to $\pm 0.2\%$ with the reported values,[7] under similar conditions.

2.3. DENSITY

Densities were measured with a vibrating tube digital densimeter (Model DMA 602/DMA 60, Anton Paar). The thermal stability, maintained by a proportional temperature controller, Tronac PTC 40, was better than $\pm 1 \times 10^{-3}$°C. The densimeter was calibrated with water (density 0.997047 g-cm^{-3} at 25°C[8]) and dry air. The reproducibility of the measurements was better than 3×10^{-6} g-cm^{-3} for dilute solutions. All the solutions were made by weight.

3. RESULTS

The integral enthalpies of solutions ΔH_s^o of monosaccharides, deoxymonosaccharides, methylglycosides, and polyols in water are given in Tables I to III. The corrections due to mutarotation, calculated from mutarotational constants[9] and heats of mutarotation,[10,11] were found to be negligible and within our experimental error. Since measurements were made at the low molalities of 10^{-5} to $10^{-3}m$ and no concentration dependence for the enthalpy of solution was observed, the average value of ΔH_s was assumed to be equal to ΔH_s^o. The uncertainties reported in the values of ΔH_s^o are the standard deviations and are the 95% confidence limit.

Calorimetric enthalpies at 25°C reported in the literature for α-D-glucose and β-D-glucose[10,12] agree well with our values. However, the enthalpies reported by Parks and Manchester[13] for meso-erythritol, mannitol, and galactitol do not show good agreement with our results. Since their values were measured at considerably higher concentrations (0.114 - 0.3634m), we believe our results to be more accurate.

No meaningful information could be drawn from partial molal enthalpy data due to the large uncertainty in the heats of formation of the pure elements needed to establish a common standard state. It is imperative to have precise and accurate data on heats of formation of such compounds, otherwise the subtle differences in the hydration properties of molecules such as mannitol and sorbitol could be masked.

The limiting heat capacity changes ΔC_p^o at 30°C, derived from ΔH_s^o at 25 and 35°C, are given in Tables I (monosaccharide and their deoxy-derivatives), II (methylglycosides), and III (polyols). The limiting partial molal heat capacities,

$$\overline{C}_{p,2}^o = \Delta C_p^o + C_{p,2}^* \quad (1)$$

where $C_{p,2}^*$ is the heat capacity of the pure solid compound, are included

Table I. Standard Thermodynamic Parameters for Monosaccharides and Their Deoxyderivatives in Aqueous Solution[a]

Compound	ΔH_s^o 25°C	ΔH_s^o 35°C	ΔC_p^o	$C_{p,2}^*$ 30°C	$\bar{C}_{p,2}^o$ 30°C	$\bar{V}_2^o$ 25°C
α-D-Glucose	10.70±0.04 10.79[b] 11.04±0.04[c]	12.04±0.12	134±12	223[d]	357 251[e] 326[f] 331[d]	111.7, 112.2±0.04[g] 111.9[h], 111.5[i]
β-D-Glucose	4.42±0.07 4.70[a] 4.54±0.27[c]	5.70±0.07	128±10	223[d]	351	111.72
D-Galactose	17.20±0.10	18.55±0.09	139±14	217[d]	356 324[c]	110.64 111.9±0.3[g] 110.7[j]
D-Mannose	6.86±0.06	8.34±0.09	148±11	215[d]	363 337[d]	111.96 111.7±0.3[g]
D-Fructose	8.80±0.10	11.20±0.09	242±13	233[d]	475 352[d]	110.88 110.4±0.4[g]
L-Sorbose	6.94±0.03	9.11±0.05	217±6	233	450	111.54 110.6±0.4[g]
D-Arabinose	13.24±0.06	14.58±0.10	134±12	184	318	94.00
D-Xylose	11.98±0.05	12.95±0.07	97±8	184	281	95.60 94.8±0.4[g]
L-Xylose	12.19±0.08	13.25±0.10	106±13	184	291	95.59
D-Lyxose	10.10±0.01	11.11±0.11	101±12	184	285	95.69
D-Ribose	13.04±0.14	14.14±0.09	110±15	184	294	95.56 95.3[j]
2-Deoxy-D-Glucose	9.43±0.06	11.23±0.15	180±16	201	381	110.4±0.5[j]
2-Deoxy-D-Galactose	10.91±0.10	12.70±0.17	180±10	201	386	
6-Deoxy-D-Galactose	1.97±0.10	3.90±0.07	193±12	201	394	
2-Deoxy-D-Ribose	10.73±0.13	12.39±0.09	169±18	164	333	93.8±0.4[g]

[a]Units for ΔH_s^o are kJ-mol^{-1}, for heat capaacities J-K^{-1}-mol^{-1}, and for volumes cm^3-mol^{-1}. [b]Ref. 17. [c]Ref. 20. [d]Ref. 28. [e]Ref. 32. [f]Ref. 30. [g]Ref. 36. [h]Ref. 37. [i]Ref. 38. [j]Ref. 10.

Table II. Standard Thermodynamic Parameters for Methylglycosides of Monosaccharides in Aqueous Solution

Pyranoside	ΔH_s^{oa} 25°C	ΔH_s^{oa} 35°C	ΔC_p^{ob} 30°C	$C_{p,2}^{*\,b}$ 30°C	$\bar{C}_{p,2}^{o\,b}$ 30°C	$\bar{V}_2^{oc}$ 25°C
1-O-Methyl-α-D-Glucopyranoside	3.67±0.07	5.67±0.03	200±10	238	438	133.29, (133.2)[e] (132.6±0.4)[d]
1-O-Methyl-β-D-Glucopyranoside	9.18±0.09	11.12±0.06	194±11	238	432	132.09, (133.6±0.5)[d] (133.5)[e]
3-O-Methyl-α-D-Glucopyranoside	8.27±0.07	10.19±0.05	192±8	238	430	133.98 (134.0±0.5)[d]
1-O-Methyl-α-D-Galactopyranoside	-1.07±0.04	0.91±0.01	198±5	238	436	(132.6±0.4)[d]
1-O-Methyl-β-D-Galactopyranoside	2.64±0.05	4.64±0.05	198±7	238	436	131.37 (132.9±0.2)[d]
1-O-Methyl-α-D-Xylpyranoside	1.82±0.05	3.50±0.04	168±7	201	369	116.36
1-O-Methyl-α-D-Mannopyranoside	9.09±0.07	11.16±0.06	207±9	238	445	132.87 (132.9±0.2)[d]
1-O-Methyl-β-D-Xylopyranoside	5.91±0.03	6.89±0.03	170±4	201	371	117.39
1-O-Methyl-β-D-Arabinopyranoside	9.58±0.06	11.27±0.06	169±8	201	370	

[a]Units:kJ-mol^{-1}. [b]Units:J-K^{-1}-mol^{-1}. [c]Units:cm^3-mol^{-1} [d]Ref. 22. [e]Ref. 4.

Table III. Standard Thermodynamic Parameters of Polyols in Aqueous Solution

compound	$\Delta H_s^{o\,a}$ 25°C	$\Delta H_s^{o\,a}$ 35°C	$\Delta C_p^{o\,b}$	$C_{p,2}^{*\,b}$ 30°C	$C_{p,2}^{o\,b}$ 30°C	$V_2^{o\,c}$ 25°C
D-Sorbitol	16.86±0.07	18.18±0.05	194±8	230	424 (423)[e]	119.26 (119.16)[e] (118.2±0.3)[f] (119.33)[g]
D-Mannitol	21.54±0.09 (22.54±0.21)[d]	23.86±0.04	232±9	229	461 (455)[e]	119.53 (119.71)[e] (118.6)[f]
D-Galactitol	29.22±0.12 (29.67)[d]	31.38±0.10	216±16	229	445	-
D-Arabitol	18.71±0.08	20.30±0.07	159±8	192	351 (375)[e]	(103.31)[d]
D-Ribitol	18.55±0.06	20.17±0.10	162±12	192	354 (376)[e]	(103.11)[e]
D-Xylitol	22.40±0.012	23.74±0.14	134±9	192	326 (346)[e]	(100.1±0.2)[e]
Meso-Erythritol	22.38±0.08 (23.31±0.04)[d]	23.61±0.18	124±21	152	276 (310)[e]	(87.10)[e]
Pentaerythritol	22.02±0.12	23.31±0.16	129±17	172	301 (329)[f]	(86.7±0.4)[f]
Myo-Inositol	14.98±0.12	16.46±0.04	148±13	221	-	-

[a]Units: kJ-mol^{-1}. [b]Units: J-K^{-1}-mol^{-1}. [c]Units: cm^3-mol^{-1}. [d]Ref. 21. [e]Ref. 29, $C_{p,2}^o$ at 298.15 K. [f]Ref. 40. [g]Ref. 39.

in these tables. $C^*_{p,2}$ for some of the compounds at various temperatures are available in the literature.[14-17] The average value of $(\partial C^*_{p,2}/\partial T)$ calculated from the reported data was used to find $C^*_{p,2}$ at 30°C. For other compounds, $C^*_{p,2}$ values were estimated from data for the heat capacity for similar pure solutes with temperature corrections applied where necessary. For sugars and *myo*-inositol, the heat capacity used for the pure solutes was 1.224 J-K^{-1}-g^{-1}; for polyols, 1.2604 J-K^{-1}-g^{-1} was used.

Comparison of $\overline{C}^o_{p,2}$ values for various polyols, with those determined by Dipaola and Belleau[18] using a flow microcalorimeter, show fairly good agreement, except for arabitol and ribitol for which their values are higher. $\overline{C}^o_{p,2}$ values for mannitol and sorbitol also match those of Bonner and Cerutti.[20] However, the value for pentaerythritol is lower than that of Jolicoeur and Lacroix.[19] $\overline{C}^o_{p,2}$ for glucose is in good agreement with that determined by Bonner and Cerutti but much higher than that of Edsall.[21] Recently Kawaizumi *et al.*[17] have reported lower values of $\overline{C}^o_{p,2}$ than ours for D-glucose, D-manose, D-galactose, and D-fructose at 30°C. The discrepancy in the case of fructose is particularly large.

The apparent molal volume ϕ_v was calculated from density data, using the equation:

$$\phi_v = M_w/d - 1000\,(d - d_o)/d\,d_o m \qquad (2)$$

where m is the molality, d is the density of solution M_w is molecular weight of the solute, and d_o is the density of water. Since the density measurements were carried out within 5-10 minutes after solution preparation, the values correspond to the conformation existing in the solid state of the compound.

The results were fitted using the least squares method to find the partial molal volumes, $\overline{V}^o_2$ which are reported in Tables I-III. Our $\overline{V}^o_2$ for monsaccharides agree with literature values[22-24], those for mannitol and sorbitol are in excellent agreement with those of Dipaola and Belleau[18] and Hoiland and Vikingstad.[25] However, Edward *et al.*[26] have reported values for both compounds that are 1 cm^3-mol^{-1} or 1% lower.

Effects of substitution, of -OH by -OCH$_3$ or -CH$_2$ and of -CH$_2$OH by -CH$_3$ in the sugar ring, on $\Delta\overline{C}^o_p$, $\overline{C}^o_{p,2}$, $\overline{V}^o_2$ values are shown in Table IV. Starting with tetrahydropyran (THP) or tetrahydropyran-alcohol (THPA) (in the case of sugars, and corresponding monohydric alcohols in the case of polyols), the effect of substitution of OH for H (obtained by dividing the total change in the thermodynamic parameter by the

Table IV. Group Substitution Effect of Some Sugars and Polyols on $\Delta\overline{C}_p^o$, $\overline{C}_{p,2}^o$ and $\overline{V}_2^o$

Group Substitution	$(\Delta\overline{C}_p^o)^a$	$(\overline{C}_{p,2}^o)^a$	$(\overline{V}_2^o)^b$
	30°C		25°C
	OH→OCH_3		
α-D-Glucose → 1-0-Methyl-α-D-Glucoside	66	76	21.59
β-D-Glucose → 1-0-Methyl-β-D-Glucoside	66	76	20.27
D-Galactose → 1-0-Methyl-α-D-Galactoside	59	65	21.96
D-Mannose → 1-0-Methyl-α-D-Mannoside	59	76	20.91
D-Xylose → 1-0-Methyl-α-D-Xyloside	71	78	20.77
	CHOH → CH_2		
α-D-Glucopyranose → 2-Deoxy-D-Glucose	46	19	-1.30
D-Galactose → 2-Deoxy-D-Galactose	46	15	-
D-Ribose → 2-Deoxy-D-Ribose	59	39	-1.76
	CH_2OH → CH_3		
D-Galactose → -Deoxy-D-Galactose	54	25	
	CHOH Addition		
Aldohexose Series	34[c]	74[c]	15 to 18
Glycosidic Series	30	66	16 to 17
Deoxy compounds of Monosaccharides	13[c]	48 to 61	16.6
Polyols	50 to 70	98 to 100	16 to 17

[a]Units: J-K^{-1}-mol^{-1}. [b]Units: cm^3-mol^{-1}. [c]Taking average of $\Delta\overline{C}_p^o$ or $\overline{C}_{p,2}^o$ for aldopentoses and aldohexoses.

number of -H atoms replaced) on $\overline{C}^{o}_{p,2}$, $\Delta\overline{C}^{o}_{p}$ and $\overline{V}^{o}_{2}$ are reported in Tables V and VI. $\Delta\overline{C}^{o}_{p}$ values for THP or THPA at 30°C were obtained by interpolating the $\Delta\overline{C}^{o}_{p}$ data reported at various temperatures by Franks *et al.*[27] $\overline{C}^{o}_{p,2}$ for other compounds were taken from the literature.[20,28,29]

4. DISCUSSION

$\Delta\overline{C}^{o}_{p}$ or $\overline{C}^{o}_{p,2}$ (Tables I-III) for all the polyhydroxy compounds studied are large and positive indicating that these polyfunctional nonelectrolyte solutes act as overall water structure-promoters.

Substitution of H by OH has been shown to result in a decrease in $\overline{C}^{o}_{p,2}$ for various homologous series of alcohols,[20,30] amines,[30] carboxylic acids[30] and amides.[30] Similar decreases in $\Delta\overline{C}^{o}_{p}$ have been reported for propane and its hydroxy derivatives[31] and for aminoacids.[32] From Tables V and VI it is interesting to note that the decrease in $\Delta\overline{C}^{o}_{p}$ and $\overline{C}^{o}_{p,2}$ is much less for polyhydroxy compounds. This indicates that the $\Delta\overline{C}^{o}_{p}$ and $\overline{C}^{o}_{p,2}$ contribution from the hydroxyl groups in these molecules is higher than that in predominantly apolar solutes. This is further supported by the observation that the $\overline{C}^{o}_{p,2}$ for polyhydroxy compounds calculated using group contribution for constituent groups[28] (obtained by using additivity relations in predominantly apolar solutes) are lower than the experimentally observed $\overline{C}^{o}_{p,2}$.

The high heat capacity observed in such compounds can be attributed to stronger or more extensive hydrogen bonding between solute hydroxyl groups and water molecules. Similar conclusions have been drawn by others from spectroscopic[33] and thermodynamic studies.[12] The stronger or more extensive hydrogen bonding between polyhydroxy nonelectrolytes and water has been explained in terms of a specific hydration model.[34] In sugars, the equatorial hydroxyl oxygens on the same side of the pyranose ring in the Cl chain conformation are 4.82A apart. This distance corresponds to the second nearest neighbor oxygen distance in water.[35] X-ray diffraction studies[36] also indicate a high concentration of water molecules at 4.9A apart. This concept of compatibility of sugar structure with tetrahedral structure of water has been invoked by various researchers[11,18,34] and our volume studies are also consistent with that view. From Tables V and VI, it can be seen that the effect on $\overline{V}^{o}_{2}$ of substitution of H by OH in corresponding monohydric alcohols or hydrocarbons to obtain sugars or polyols is negligible confirming that there is little perturbation in water structure on hydroxyl group substitution.

Table V. Effects of Substitution of H by OH on Heat Capacities and Partial Molal Volumes in Transforming Monohydric Alcohols or Hydrocarbons to the Corresponding Sugars

Compounds	$(\Delta\overline{C}_p^o)^a$	$(\Delta\overline{C}_p^o)$ per OH^a	$(\overline{V}_2^o)^b$	$(\overline{V}_2^o)$ per OH^b
THPA[c] Sorbose	217-115[e]=102	25.5	111.54-108.1[f] =3.44	0.86
THPA Fructose	242-115[e]=127	31.75	110.88-108.1[f] =2.78	0.69
THPA	134-115[e]=19	4.75	111.70-108.1[f] =3.60	0.90
Glucose(α β)	128-115[e]=13	3.25	111.72-108.1[f] =3.62	0.90
THPA Galactose	139-115[e]=24	6	110.64-108.1[f] =2.54	0.63
THPA Mannose	148-115[e]=33	8.25	110.96-108.1[f] =2.86	0.72
THP[d] Arabinose	134-156[e]=-22	-5.5	94.0-90.6[f] =3.4	0.85
THP Lyxose	101-156[e]=-55	-13.75	95.69-90.6[f] =5.09	1.27
THP Xylose	97-156[e]=-59	-14.75	95.60-90.6[f] =5.0	1.25
THP Ribose	97-156[e]=-59	-14.75	95.56-90.6[f] =4.96	1.24
THPA 2-Deoxyglucose	180-115[e]=65	22	110.4[g]-108.1[f] =2.3	0.76
THPA 2-Deoxygalactose	185-115[e]=70	23		-
THPA 6-Deoxygalactose	193-115[e]=78	26		-
THP 2-Deoxyribose	169-156[e]=13	6.5	93.8[g]-90.6[f] =3.2	1.6

[a]Units: J-K^{-1}-mol^{-1}. [b]Units: cm^3-mol^{-1}. [c]THPA = tetrahydropyran-2-dimethanol. [d]THP = tetrahydropyran. [e]Ref. 27. [f]Ref. 4. [g]Ref. 22.

Franks *et al.*,[4] from their partial molal volume studies have pointed out that the β-hydroxyl of sugars is more hydrated than the α-hydroxyl group. Our results on partial molal volumes of various anomers do not show such a pattern. Also, $\Delta\overline{C}_p^o$ or $\overline{C}_{p,2}^o$ results for α,β-glucose (Table I) and a series of methylglycosides (Table II) do not show any difference for the two anomers. From our results it would appear that a change of orientation of the -OH or -OCH_3 group from α to β does not result in any change in limiting thermodynamic properties. However, this observation is not consistent with the compatibility concept invoked earlier.

Furthermore $\Delta\overline{C}_p^o$ and $\overline{C}_{p,2}^o$ for stereoisomers, within a particular series of compounds (Tables I-III) are the same (the small difference falling within experimental error) indicating similar hydration behavior for the two stereoisomers. Similar observations for glucose, mannose and galactose have been made by others from spectroscopic[34] and thermodynamic studies.[17]

1-O-methyl-glucopyranoside has the same $\Delta\overline{C}_p^o$ or $\overline{C}_{p,2}^o$ as 3-O-methyl-glucopyranoside (Table II) and 2-deoxy-D-glucose the same as 6-deoxy-D-glucose (Table I). It appears, therefore, that the position of the -OCH_3 or CH_2 groups does not affect the mode or extent of hydration.

The compounds with the 1C pyranose chain conformation in the crystalline state appear to have higher $\overline{C}_{p,2}^o$ than those with the C1 pyranose chain conformation *e.g.* the $\overline{C}_{p,2}^o$ for D-arabinose with 1C structure (see Table I) is larger than that for its stereoisomers with C1 structure. It cannont be said, at the moment, that the higher heat capacity for L-sorbose in comparison to that for D-glucose is due to the difference in the conformation (C1 and 1C) or due to functional groups. However, Kawaizumi[17] has also reported higher $\overline{C}_{p,2}^o$ for fructose.

D- and L-stereoisomers do not show any difference in the limiting thermodynamic properties as seen for D-xylose and L-xylose, both having the same $\Delta\overline{C}_p^o$ and $\overline{V}_2^o$ (Table I). Similar conclusions have been drawn[18] from thermodynamic studies of D- and L-arabitol.

Comparison of the results in Table IV shows that the substitution of -OH by -OCH_3 results in an increase of about 66 J-K^{-1}-mol^{-1} in ΔC_p^o, 76 J-K^{-1}-mol^{-1} in $C_{p,2}^o$, and 20$\pm$1 cm^3-mol^{-1} in $\overline{V}_2^o$. The increase in ΔC_p^o agrees with the value obtained for CH_2 addition in predominantly apolar compounds. But the change in $\overline{C}_{p,2}^o$ (76 J-K^{-1}-mol^{-1}) is less than that observed in apolar solutes (88 J-K^{-1}-mol^{-1}) indicating that the change in $C_{p,2}^*$ accompanying CH_2 addition in polyhydroxy compounds

Table VI. Effects of Substitution of H by OH on Heat Capacities and Partial Molal Volumes in Transforming Monohydric Alcohols to the Corresponding Polyol

Alcohol Polyol	$(\overline{C}^{o}_{p,2})^{a}$ 25°C	$(\overline{C}^{o}_{p,2})^{a}$ per OH	$(\overline{V}^{o}_{2})^{b}$ 25°C	$(\overline{V}^{o}_{2})^{b}$ per OH
n-Propanol Glycerol	253[c](240.2)[d]- 352.9[c] =-99.9(-112.7)	-49.9 (-56.3)	070.95-70.75 =0.20	0.10
n-Butanol *meso*-Erythritol	310[d]-437[c] =-127	-31.7	087.10[d]-86.62[e] =0.48	0.16
n-Pentanol Ribitol	375.6[d]-523.8 =-148.2	37.05	103.11[d]-102.62[e] =0.49	0.12
n-Pentanol Xylitol	346.4[d]-523.8[c] =-177.4	-44.3	102.14[d]-102.62[e] =-0.48	0.12
n-Pentanol Arabitol	374.6[d]-523.8[c] =-149.2	-37.2	103.31[d]-102.62[e] =0.68	0.17
n-Hexanol Sorbitol	423[d]-611[c] =-188	-37.6	119.26-118.65[e] =0.65	0.12
n-Hexanol Mannitol	455[d]-611 =-156	-31.2	119.53-118.65[e] =0.88	0.17
n-Hexanol Galactitol	-	-	-	-
neo-Pentanol Penta Erythritol	328.8[c]-503.5[c] =-174.7	-58.2	-	-

[a]Units: J-K^{-1}-mol^{-1}. [b]Units: cm^{3}-mol^{-1}. [c]Ref. 29. [d]Ref. 18. [e]Ref. 25.

is lower by about 12 J-K^{-1}-mol^{-1} than that observed in hydrophobic solutes. Comparison of $C^{*}_{p,2}$ for alkoxy ethanols and dialkoxy ethanes[37] also shows that for the OH to CH_3 group substitution the change in $C^{*}_{p,2}$ is 7-18 J-K^{-1}-mol^{-1} less than that accompanying CH_2 addition in predominantly apolar solutes.

In predominantly apolar solutes, the addition of a -CHOH group

results in either a decrease or no change in the heat capacity whereas the partial molal volume increases by 16-18 cm^3-mol^{-1}.[20,25] Similar volume changes (15-18 cm^3-mol^{-1}) per -CHOH addition are observed in polyhydroxy solutes but the heat capacities show an increase, as seen in Table IV. This increase in heat capacity reflects the positive contribution of the -OH group towards heat capacity in such solutes.

Within the polyol series, mannitol and galactitiol have higher heat capacities than sorbitol. Also, arabitol and ribitol are higher than xylitol. These differences have been explained by Dipaola and Belleau[18] in terms of compatibility between polyol conformation and water structure. NMR studies[38] show that sorbitol, xylitol, and ribitol possess nonplanar sickle conformation in aqueous solution, whereas all other polyols mentioned above have planar zigzag conformation. With ribitol as an exception, it can be seen that a planar zigzag structure, being more compatible with the water structure, shows higher ΔC_p^o or $C_{p,2}^o$ than a nonplanar structure.

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